

A Label–Score Alignment Pitfall in SNR-Conditioned OOD Detection for Single-Channel Blind Source Separation

Bin Nong, Weihong Fu, and Zhuoyun Jiang

Abstract—Evaluating out-of-distribution (OOD) detection for communication signals is routinely conditioned on the operating SNR. We show this conditioning is fragile to label misalignment. Storing per-source SNR labels interleaved (`np.repeat`) while the scores they annotate are stacked slot-wise (`np.tile`) manufactures a spurious phenomenon in which two OOD scorer families appear complementary across SNR and a training-free SNR-routed ensemble appears to lift AUROC from 0.50 to 0.625 ± 0.031 . We formalize the failure as an indexing-permutation confound and prove that whenever OOD scores drift with SNR, *any* label permutation that alters the per-bin SNR composition converts that drift into spurious per-SNR separability (routed weighted AUROC up to 0.746). The same pipeline is then shown to hide a second, independent misalignment—per-source labels swapped alongside permutation-invariant alignment—with the opposite effect: a genuine same-mixture detection margin (AUROC 0.54) attenuated to an apparent null. Randomness-oriented safeguards (multi-seed averaging, significance tests, robustness batteries) detect neither error. Doubly corrected, no post-hoc scorer gives deployable *pooled* discrimination (routed 0.508 ± 0.026 ; five seeds, four independent test realizations); a modest *same-mixture* embedding margin survives (prototype AUROC 0.54, 0.60–0.65 at $\text{SNR} \geq 5\text{ dB}$), logit scorers are inverted, and the margin collapses when the unknown source is the weak one.

Index Terms—Blind source separation, open-set recognition, out-of-distribution detection, evaluation methodology, label misalignment.

I. INTRODUCTION

SINGLE-CHANNEL blind source separation (SC-BSS) of co-frequency communication signals is increasingly paired with per-source modulation identification [1], [2], and a deployed receiver must additionally *reject* sources whose modulation it has never seen—an open-set recognition problem [3], [4] for which post-hoc OOD scorers of learned embeddings and logits [5], [6], [7], [8], [9] are the standard toolkit. Because OOD score distributions drift with noise, communication-signal evaluation is routinely conditioned on the operating SNR—per-SNR-bin profiles and weighted averages [14], [15], [4].

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This letter shows that the conditioning step itself is a single point of failure. A one-line labeling-convention mismatch misaligns every known-pool score with the wrong SNR bin, and each “per-SNR bin” then compares known and unknown pools at *different* true SNRs: the bins measure SNR separability, not OOD separability. The artifact is insidious precisely because it is *systematic*: multi-seed averaging, significance testing, and robustness batteries are randomness-oriented safeguards, and they cannot detect deterministic pipeline errors [11], [12]. We contribute: (i) a formalization of the failure as an indexing-permutation confound, including a proposition giving the exact conditions (SNR-dependent scores plus a composition-changing label permutation) under which *any* such misalignment manufactures spurious per-SNR gains (routed weighted AUROC up to 0.746), and why the trusted checks fail to catch it; (ii) a deployment-faithful evaluation protocol (verified alignment, strict train/reference/test separation, unknown-free calibration, operating-point metrics, protocol-wise and same-mixture comparisons, independent test realizations)—whose same-mixture control exposes a *second*, independent label misalignment in the same pipeline (per-source labels swapped alongside permutation-invariant alignment), with the opposite sign: it attenuates a genuine margin to an apparent null; and (iii) the doubly corrected study on an open-set SC-BSS benchmark (six scorers, five seeds, two backbones, four independent test sets, LOMO splits, per-class near/far-OOD breakdowns, a calibrated operating point, a controlled interference sweep): no deployable *pooled* discrimination, but a modest *same-mixture* embedding margin that is inverted for logit scorers and collapses when the unknown source is weak.

II. OPEN-SET SC-BSS SETUP

Signal model and SNR definition. We study the two-source co-frequency mixture $y(t) = \alpha_1 s_1(t) + \alpha_2 s_2(t) + n(t)$ at complex baseband. Each s_i is a unit-power synthetic waveform (16 kHz sampling, 256 symbols at 16 samples/symbol, root-raised-cosine pulse shaping with roll-off 0.35, 3-tap complex Gaussian multipath normalized to unit gain, carriers 2000 Hz/2005 Hz with ± 5 Hz uniform jitter, $T = 4096$ samples [14]), mixed with $\alpha_1 = \alpha$, $\alpha_2 = 1 - \alpha$, $\alpha \sim U(0.4, 0.6)$, plus AWGN. Throughout, SNR denotes the *mixture* SNR $\text{SNR}_{\text{mix}} = 10 \log_{10}(\mathbb{E}|\alpha_1 s_1 + \alpha_2 s_2|^2 / \mathbb{E}|n|^2)$. Since the sources are unit-power, the per-source SNR of source i is $\text{SNR}_i = \text{SNR}_{\text{mix}} + 10 \log_{10}(\alpha_i^2 / (\alpha_1^2 + \alpha_2^2)) \approx \text{SNR}_{\text{mix}} - 3\text{ dB}$ under near-equal mixing, and the SIR $20 \log_{10}(\alpha_1 / \alpha_2)$ sweeps

$[-3.5, 3.5]$ dB with the α draw. The per-source label stored by the evaluation pipeline is this shared SNR_{mix} .

Modulations, protocols, pools. Modulations split into known $\mathcal{K} = \{\text{BPSK}, \text{QPSK}, \text{8PSK}, \text{16QAM}\}$ (training only) and held-out $\mathcal{U} = \{64\text{QAM}, \pi/4\text{-DQPSK}, \text{MSK}, \text{OFDM-QPSK}\}$; test protocols are kk/ku/uu by the number of unknown sources (deterministic seed 99999; 192 kk, 192 ku, and 96 uu mixtures per bin of the grid $\{-10, -5, \dots, 20\}$ dB). The *known pool* holds both slots of the kk mixtures and the *unknown pool* the OOD slot of each ku mixture plus both slots of uu (2688 sources each; 384 per SNR bin). The separator is a lightweight complex-valued CNN with complex squeeze-and-excitation (C-SE) attention ($\sim 69\text{K}$ parameters) plus a shared per-source modulation head (64-d embedding z , 4 logits f), trained with permutation-invariant SI-SDR + cross-entropy [16], [17] on known modulations only. Closed-set reference points (kk): SI-SDR -0.89 dB pooled; truth-anchored modulation accuracy 0.52—an interference-limited regime ($\text{SIR} \approx 0$ dB), the stress test under which per-source OOD detection would have to operate.

Scorers and leakage control. Six post-hoc OOD scorers (larger = more OOD) are applied per source: Energy $-\log \sum_k e^{f_k}$ [7], MSP [5], ODIN [6], Mahalanobis [8], nearest-prototype distance, and a VOS-inspired (training-free, post-hoc) virtual-outlier distance [9], all at literature defaults fixed *a priori* (ODIN $T=1000$, $\varepsilon=0.005$; Mahalanobis shared covariance, shrinkage 0.1; VOS-inspired scale 2, 100 outliers/class); every fitted quantity (prototypes, covariance, virtual outliers, thresholds) is estimated on a disjoint reference pool (seed 88888, kk protocol), and **no unknown-class sample is used for any fitting, selection, or calibration step**—the routing boundary (0 dB) is a development-time choice, fixed from the validation-profile crossover of the original study and frozen before any corrected-test evaluation, and the LOMO study selects boundaries exclusively on a further pseudo-ODD validation split (seed 77777). The rejection threshold τ is the 95th percentile of known-pool reference scores (5% false-rejection target). The *a-priori SNR-routed* score uses Energy at $\text{SNR} \leq 0$ dB and Prototype otherwise.

III. THE PITFALL

Mechanism and formalization. Index the test mixtures $m = 1, \dots, N$ in SNR-block order. The evaluation stores per-source scores q_j stacked *tile-wise* and per-source SNR labels ℓ_j written *interleaved* (`np.repeat`), so score j belongs to mixture $\sigma_q(j) = ((j-1) \bmod N) + 1$ but is paired with the label of mixture $\sigma_\ell(j) = \lceil j/2 \rceil$: an indexing-permutation mismatch $\sigma_q \neq \sigma_\ell$ (Fig. 2). Binning by ℓ assigns to stored bin b known scores whose *true* SNRs differ from b wherever the permutations disagree; the exactly labeled unknown pool is unaffected, so each per-bin AUROC measures the score’s *SNR separability*, not OOD separability, with a magnitude set by the deterministic label-truth SNR gap (up to 15 dB here). A one-line audit exposes the corruption: binning the known pool’s stored *modulation* labels by the stored SNR labels gives impossible per-bin class counts (114/96/100/74; the protocol contributes exactly 96 mixtures per known modulation per true bin); the corrected (tile) labels recover 96/96/96/96.

A conditional proposition. The artifact instantiates a general confound whose conditions can be stated exactly.

Proposition 1. *Let S be an OOD score that is conditioned on the true SNR C but carries no OOD information beyond it, $S \perp O \mid C$ for the OOD label $O \in \{k, u\}$, and whose family $\{p(s \mid c)\}$ is stochastically ordered in c . If the stored labels assign to bin b a known pool whose true-SNR distribution $P(C_k \mid \ell_k=b)$ has mass at $c \neq b$, while the exactly labeled unknown pool sits at b , then the per-bin AUROC $A_b = P(S_u > S_k \mid \ell_u=\ell_k=b) \neq \frac{1}{2}$, with sign set by $b - \mathbb{E}[C_k \mid \ell_k=b]$.*

Sketch. Marginalizing the corrupted known labels, $A_b = \int P(S_u > S_k \mid c_u=b, c_k) dP(c_k \mid \ell_k=b)$; under $S \perp O \mid C$ the integrand equals $\frac{1}{2}$ at $c_k = b$ and moves monotonically with $b - c_k$ by stochastic ordering, so any mass away from b biases A_b . $\square$

The conditions are exact: the score must drift with SNR (all six scorers here do) and the permutation must change the per-bin SNR composition (block-aligned shifts are exact no-ops, below).

Generality: permutation study. The repeat-vs-tile bug is one member of the class delimited by Proposition 1. Re-binning the *corrected* scores with synthetic label permutations π over the known pool (5-seed mean weighted AUROC of the *a-priori* routed score): $\pi_{\text{identity}} = 0.508$; π_{repeat} (the original bug, isolated on clean pools) = 0.619; $\pi_{\text{reverse}} = 0.746$; ten uniform *random* permutations = 0.632 ± 0.005 ; block-structure preserving shifts are exact no-ops (0.508). Per-bin AUROCs reach 0.85 under π_{reverse} . Two confounds combine: mismatched known bins separate by their true-SNR composition (under π_{random} each bin approximates “full known pool vs unknown at b ”, exposing opposite monotone SNR gradients for Energy and Prototype), and the fixed *a-priori* routing rule harvests both gradients by construction. Notably, single-scorer weighted averages stay near 0.48–0.51 under every π : the average can look innocent while the per-bin structure is entirely spurious—which is exactly how the artifact presented as *complementarity* rather than as a global lift. Label-index alignment is therefore not an implementation detail but a confound-control condition for conditioned evaluation.

The manufactured phenomenon. Under the buggy labels the benchmark presented a perfect complementarity (Prototype 0.84 AUROC at 10 dB, collapsing below 0 dB; Energy 0.74 at -5 dB), and routing on it appeared to lift the pair-weighted AUROC from 0.50 to 0.625 ± 0.031 (five seeds) against a post-hoc oracle of 0.681 (Fig. 1a). It also produced a full supporting narrative: “graceful degradation” under simulated SNR-estimation error, “robustness” to SIR/timing/carrier perturbations, “consistent” ablations.

Why it survives the standard checks. The misalignment is a deterministic function of the data layout, identical for every seed and condition: multi-seed averaging *confirms* it (± 0.031), and significance tests are significant precisely because the SNR-mismatch signal is real—what is wrong is the bins’ interpretation. Robustness perturbations act on the signals, not the label geometry, so each pass reads as independent evidence; and the pattern is mechanistically narratable (“confidence inverts at high SNR”), which is how it survives internal

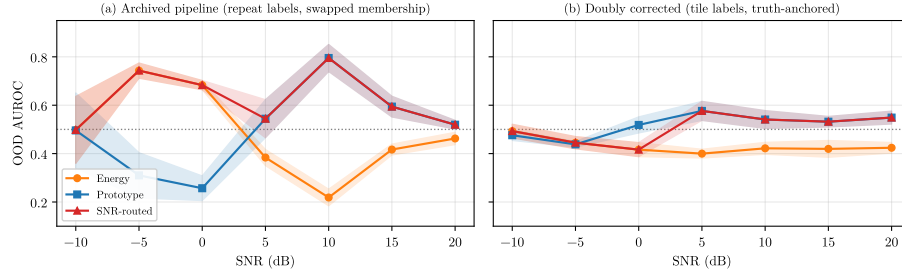


Fig. 1. Per-SNR OOD AUROC (5-seed mean $\pm$ std) under (a) the archived pipeline (repeat SNR labels, swapped pool membership) and (b) the doubly corrected evaluation (tile labels, truth-anchored). The apparent complementarity—Energy peaking at low SNR, Prototype at high SNR—is manufactured by the label misalignments; corrected, the routed profile is flat near chance (dotted line).

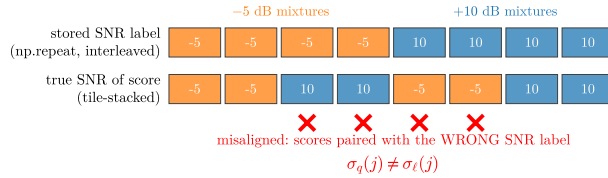


Fig. 2. The misalignment. Scores are stacked tile-wise (all slot-1 entries, then all slot-2), so their true SNR sequence alternates blocks; the stored labels repeat each mixture's SNR interleaved. Entries past the slot boundary (red crosses) are paired with the wrong SNR (up to 15 dB)—an indexing-permutation mismatch $\sigma_q(j) \neq \sigma_\ell(j)$ (Sec. III).

scrutiny. The critical check is label–score alignment against regenerated data, which standard seed- and robustness-based validation does not capture.

How it was caught. Replacing the oracle SNR in the router with a real blind SNR estimator (median-eigenvalue subspace estimator, 0.51 dB overall RMSE; classical M2M4 is inadequate here [13]) required pairing waveform-derived estimates with stored scores; the alignment verification written for that purpose failed immediately and exposed the repeat-vs-tile mismatch.

IV. CORRECTED PROTOCOL AND RESULTS

Protocol. (i) **Verified label alignment:** every dumped label array is checked against regenerated data (sequence equality, per-bin counts, per-bin label multisets). (ii) **Strict separation:** training, the reference pool (seed 88888), and the test set (seed 99999) never mix; all fitted OOD quantities come from the reference pool. (iii) **Unknown-free calibration:** τ is the 95th percentile of known-pool reference scores; no unknown-class data is used for any fitted quantity, and the routing boundary is frozen from development (Sec. II). (iv) **Operating-point metrics**—Det@ τ , FPR95 (known-pool rejection rate at 95% unknown detection), OSCR [10], joint known/unknown accuracy—alongside per-SNR and pooled AUROC. (v) **Protocol decomposition:** pooled AUROC could confound modulation novelty with mixture composition (the known pool draws only from kk mixtures); we report protocol-wise AUROCs and a *same-mixture* comparison (known vs unknown slot within each ku mixture), which removes the confound by construction (uu mixtures contain no known source and contribute no such control). (vi) **Independent**

TABLE I
WEIGHTED-AVERAGE PER-SNR AUROC, ARTIFACT VS CORRECTED LABELS (5 SEEDS, MEAN $\pm$ SEED STD); CORRECTED POOLED AUROC ON THE CONCATENATED 5-SEED SAMPLES (13440 + 13440) WITH ITS OWN 95% BOOTSTRAP CI; Δ = AUROC $-$ 0.5; p FROM A TWO-SIDED LABEL-PERMUTATION TEST ($B = 10^4$). CORRECTED COLUMNS ARE DOUBLY CORRECTED (TILE SNR LABELS, TRUTH-ANCHORED MEMBERSHIP AND REFERENCE FIT); THE ARTIFACT COLUMN REPRODUCES THE ARCHIVED MISLABELED DUMPS. THE ORACLE'S MARGIN OVER ROUTED (0.042) SPLITS INTO PER-BIN MAX-SELECTION OPTIMISM (NULL 0.518 [0.513, 0.523]) PLUS THE SMALL GENUINE EMBEDDING MARGIN ($p = 5 \times 10^{-4}$ VS THE NULL).

Scorer	Artifact wavg	Corrected wavg	Corrected pooled [95% CI], Δ (p)
Energy	0.487 $\pm$ 0.032	0.431 $\pm$ 0.023	0.450 [0.443, 0.456], -0.050 (10^{-4})
MSP	0.435 $\pm$ 0.012	0.394 $\pm$ 0.007	0.405 [0.398, 0.411], -0.095 (10^{-4})
ODIN	0.495 $\pm$ 0.030	0.435 $\pm$ 0.021	0.460 [0.453, 0.466], -0.040 (10^{-4})
Mahalanobis	0.526 $\pm$ 0.043	0.529 $\pm$ 0.056	0.525 [0.517, 0.531], $+0.025$ (10^{-4})
Prototype	0.502 $\pm$ 0.008	0.518 $\pm$ 0.022	0.509 [0.502, 0.516], $+0.009$ (0.008)
VOS-inspired	0.502 $\pm$ 0.008	0.518 $\pm$ 0.022	0.509 [0.502, 0.516], $+0.009$ (0.01)
SNR-routed	0.625 $\pm$ 0.031	0.508 $\pm$ 0.026	—
Oracle (6, post-hoc)	0.681 $\pm$ 0.039	0.550 $\pm$ 0.041	—

test realizations: beyond the deterministic test seed 99999, the full protocol is repeated on three further independent test sets (seeds 100001–100003). Seed standard deviations describe training-run variability; bootstrap CIs describe test-sample variability.

A second misalignment, exposed by the same-mixture control. The same-mixture comparison required by protocol item (v) contradicted the pooled corrected numbers; the audit traced it to a second label misalignment in the dump stage of the original pipeline: permutation-invariant (PIT) alignment anchored the waveforms, embeddings, and logits to the true sources but swapped the per-source *labels* (modulation indices and OOD flags) with the output slots. At SIR $\approx$ 0 dB the swap rate is $\approx$ 0.50, so the ku-derived pools were about half contaminated and every ku-dependent AUROC was attenuated toward chance (closed-set accuracy recovers from 0.42 to 0.52 when truth-anchored); the kk/uu pools, the reference pool, and the SNR labels are unaffected, as is Sec. III's analysis in kind. The failure class is Proposition 1's—deterministic label misalignment—with the opposite sign: it *erases* structure. **All corrected numbers below are truth-anchored.**

Corrected results. Under verified, truth-anchored labels no scorer provides practically useful *pooled* discrimination (Fig. 1b; Table I): the routed ensemble reaches 0.508 ± 0.026

TABLE II

OPERATING-POINT METRICS OF THE ROUTED SCORE AT THE CALIBRATED 5%-FRR THRESHOLD τ (TRUTH-ANCHORED LABELS, 5 SEEDS, MEAN $\pm$ STD). A: SPLIT-HALF REFERENCE; B: DISJOINT TRUTH-ANCHORED REFERENCE POOL (SEED 88888), REFITTED PROTOTYPES.

Metric	Variant A	Variant B
Det@ τ (unknown detection)	0.066 ± 0.040	0.064 ± 0.040
False-rejection rate (target 0.05)	0.052 ± 0.000	0.049 ± 0.002
FPR95	0.947 ± 0.009	0.948 ± 0.008
OSCR [10]	0.272 ± 0.035	0.273 ± 0.034
Joint known/unknown accuracy	0.206 ± 0.043	0.278 ± 0.045

and stays flat across three further independent test realizations (seeds 100001–100003, routed 0.501–0.517). The deviations from chance are structured: the logit scorers sit *below* chance (MSP $\Delta = -0.095$; Energy $\Delta = -0.050$; permutation $p = 10^{-4}$)—the known-class head is confidently wrong on unknown modulations—while the embedding scorers sit marginally above (Mahalanobis $\Delta = +0.025$; Prototype/VOS $\Delta = +0.009$), driven by far-ODD classes and high-SNR bins (below). The post-hoc oracle (0.550) exceeds its correlation-preserving permutation null (0.518, $p = 5 \times 10^{-4}$)—a bound, not a deployable score (Table I). Under truth-anchored labels the intervention battery leaves the routed score near chance: SIR/timing/CFO 0.488–0.504, carrier gaps ≤ 100 Hz 0.503–0.513 (500 Hz: 0.449—large separation breaks the co-frequency embedding), center loss ≈ 0.50 , embedding width 0.42–0.54; simulated SNR-estimation noise ($\sigma \leq 6$ dB) leaves routing unchanged. LOMO splits (boundary selected only on the pseudo-ODD validation split) give routed 0.482 ± 0.032 , and no split finds a genuine crossover (6/12 fall back to 0 dB, 6/12 to all-Energy)—there is no crossover to find. A second backbone without the C-SE blocks shows the same structure (routed 0.529 ± 0.021 , Mahalanobis 0.61): the margin is not backbone-specific.

Protocol decomposition. Restricting the pooled comparison by protocol (Table III) shows that pool construction is not the driver: the cross-mixture contrasts stay near chance, while the *same-mixture* contrast—known vs unknown slot within each ku mixture—exposes a real but modest margin for the embedding scorers (Prototype/VOS 0.54, reaching 0.60–0.65 at SNR ≥ 5 dB) and a strong *inversion* for the logit scorers (Energy 0.33).

Operating point. At the calibrated single threshold (Table II) the detector rejects 6.4% of unknowns at 4.9% false rejection—a near-useless trade confirmed by FPR95 (0.948), OSCR (0.273) and joint accuracy (0.278). Per SNR, Det@ τ is *exactly* 0 at ≤ 0 dB and rises to 0.16 at 20 dB (false-rejection tracking it, $0 \rightarrow 0.126$): the threshold tracks the known pool’s SNR-driven score drift and the unknown pool drifts identically. The failure is distributional, not a threshold-choice artifact.

Near vs far OOD. Per class (corrected weighted-average AUROC): embedding-distance scorers score the far-ODD families higher (Mahalanobis/Prototype: 0.58/0.58 and 0.54/0.57 on MSK/OFDM-QPSK) than the near-ODD ones (0.46–0.50 on 64QAM/ $\pi/4$ -DQPSK), while logit scorers *invert* the ordering—MSP reaches 0.609 on near-ODD $\pi/4$ -DQPSK yet collapses to 0.216 on OFDM-QPSK: the multicarrier wave-

TABLE III

PROTOCOL-DECOMPOSED WEIGHTED-AVERAGE AUROC (TRUTH-ANCHORED, 5-SEED MEAN); THE SAME-MIXTURE ROW COMPARES THE TWO SLOTS WITHIN EACH KU MIXTURE (UU HAS NO KNOWN SLOT). ODIN/VOS BEHAVE LIKE ENERGY/PROTOTYPE.

Contrast	Ener.	MSP	Mah.	Prot.	Rout.
K(kk) vs U(ku)	0.393	0.373	0.526	0.524	0.506
K(kk) vs U(uu)	0.470	0.415	0.532	0.512	0.509
K(ku) vs U(ku), same mix.	0.334	0.347	0.517	0.543	0.523

form looks *more* in-distribution to the known-class head than the known pool itself. The near/far intuition does not transfer to post-hoc scoring of separated sources.

Attribution. Three controlled observations bound the explanation. (1) Separation quality is class-independent: SI-SDR of known vs unknown sources is -0.89 vs -1.00 dB pooled (within 0.21 dB per bin)—the separator does not distort unknown modulations more than known ones. (2) The margin tracks the weaker source’s separability: sweeping the SIR at fixed 10 dB mixture SNR, the same-mixture prototype margin peaks near equal power (0.63 at SIR 0/+5 dB) and degrades or destabilizes when either source dominates (≤ 0.55 with ± 0.2 seed spread at $|\text{SIR}| \geq 10$ dB). (3) A partial margin exists without interference: on clean single-source inputs (a probe, not a deployed configuration), Mahalanobis reaches 0.660 ± 0.046 (0.77 at 10 dB, 0.82 on far-ODD OFDM) while logit scorers stay at or below chance (0.42–0.44). Together, the observed results suggest that the exploitable known/unknown margin *operating on separated sources* is real but modest, visible only against a same-mixture reference, and bounded by the weaker source’s residual interference; the logit pathway never carries one. Scope: these conclusions hold for the evaluated synthetic benchmark and protocol; dedicated open-set modulation recognition [4] with unknown-aware training is a separate, harder problem.

V. CONCLUSION

A repeat-vs-tile labeling mismatch manufactured an entire SNR-conditioned phenomenon that survived every randomness-oriented check; a PIT label swap in the same pipeline did the opposite, erasing a genuine same-mixture margin. The failure is a general indexing confound, not a one-off bug: randomness-oriented safeguards cannot detect deterministic pipeline errors. Doubly corrected, pooled post-hoc OOD detection from separation embeddings remains non-deployable on the evaluated benchmark; a modest same-mixture embedding margin survives as an honest starting point. We recommend: (1) *align labels against regenerated data* (sequence, per-bin counts, label multisets) before trusting any conditioned metric, and *audit label anchoring after every permutation/alignment step* (closed-set accuracy is the zero-cost canary); (2) *multi-seed averaging is not a pipeline check*; (3) *separate train/reference/test strictly*, calibrate without unknown-class data, and decompose pooled metrics by protocol (same-mixture included); (4) *calibrate post-hoc selection against a label-permutation null*. Code and dumps: https://github.com/BinNong/single_channel_source_seperation (release tag paper3-spl).

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